Dense-Weak Ship Detection Based on Foreground-Guided Background Generation Network in SAR Images

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***Abstract*— Currently, ship detection based on synthetic aperture radar (SAR) images still faces significant challenges, particularly in detecting weak and densely distributed ships within complex backgrounds. In areas such as ports and land, the complex background features often resemble those of densely distributed ships, leading to reduced detection accuracy. In addition, the overlapping and mutual interference of features among dense ships can cause the network to miss detections or produce false positives. Therefore, this article proposes a foreground-guided background generation network (FGBG-Net), which includes a Gaussian foreground localization (GFL) model and a background feature removal (BFR) module. The GFL module identifies the approximate high-probability regions of ship foregrounds on the feature map, guiding the network to focus on these regions. The BFR module then progressively removes background interference features based on the positions provided by the GFL module, generating feature maps that are more suitable for detecting weak and dense ships. Our network has been validated on multiple SAR ship datasets, and the experimental results demonstrate noticeable performance improvements, with a mean average precision (mAP) increase of 3.4% on the SSDD and HRSID datasets. The relevant code is available at the following link: https://github.com/Xidian-AIGroup190726/FBGBNet/tree/master**

***Index Terms*— Backgrounds removal, deep learning, Gaussian mixture model (GMM), object detection, ship detection, synthetic aperture radar (SAR).**

I. Introduction

IN RECENT years, ship detection [1], [2], [3] has become increasingly widely used in military and civilian fields. Due to the special imaging principles of synthetic aperture radar (SAR) images [4], [5], [6], it is challenging to detect weak-ships inshore and in dense arrangements.

Currently, object detectors are roughly divided into anchor-based [7], [8] and anchor-free [9] methods. These networks are widely used for ship detection in SAR images. Among them, anchor-based methods, represented by convolutional neural networks (CNN) networks [10], [11], [12], have led to many excellent designs, such as Fast-region-based convolutional neural network (RCNN) [13], Faster-RCNN [14], and Cascade-RCNN [15]. These networks use predefined anchor boxes for ship detection, while anchor-free methods, such as fully convolutional one-stage object detection (FCOS) [9], primarily rely on feature points for detection, with bounding boxes dynamically adjusted during training rather than being predefined. Recently, balanced feature-enhanced attention (BFEA) [16], an SAR ship detection model based on an attention mechanism and multiscale feature fusion, has demonstrated remarkable performance by leveraging attention mechanisms to enhance feature representation and improve detection accuracy, particularly for medium and large ships. However, whether anchor-based or anchor-free, the detection of dense-weak ships [17] in complex backgrounds [18] based on SAR images remains a significant challenge.

Unlike traditional optical images, SAR images are typically presented as single-channel grayscale images [19], resulting in less semantic richness compared to optical images. From Fig. 1, it can be seen that in complex backgrounds, the feature response of the baseline method is relatively scattered, which can easily lead to misjudgment due to background interference. Dense ships are also affected, resulting in inaccurate feature extraction. In simple backgrounds, the accuracy of feature extraction is significantly improved. For example, the Cascade-RCNN network designed by Cai and Vasconcelos [15] uses cascaded convolutions and achieves good detection accuracy, but its average precision (AP) [20] for inshore detection is much lower than for offshore detection. For anchor-free networks [21], [22], such as FCOS [9], ship detection is performed using feature points [23]. Although these networks have lightweight structures, their detection performance can still be affected by background interference.

To address the challenges of ship detection in SAR images with complex backgrounds, recent advancements include the Deformable-Detection Transformer (DETR) [24] network, which combines deformable convolution with the DETR [25] framework. Compared to traditional CNN convolutions [26], [27], deformable convolution [28] is more adaptable to complex backgrounds in SAR images, effectively integrating ship features with their contextual neighborhoods. The YOLOX series [29], [30] and similar networks have shown strong adaptability to ship detection across different scales in SAR images. In addition, Yu et al. [31] proposed a lightweight attention enhanced path aggregation feature pyramid network

Received 15 October 2024; revised 18 February 2025 and 16 April 2025; accepted 17 May 2025. Date of publication 21 May 2025; date of current version 3 June 2025. This work was supported in part by the Joint Funds of the National Natural Science Foundation of China under Grant U22B2054; in part by the National Natural Science Foundation of China under Grant 62076192, Grant 62276199, Grant 62431020, and Grant 62276201; in part by the 111 Project; in part by the Program for Cheung Kong Scholars and Innovative Research Team in University under Grant IRT15R53; and in part by the Science and Technology Innovation Project from the Chinese Ministry of Education. *(Corresponding author: Hao Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3572095

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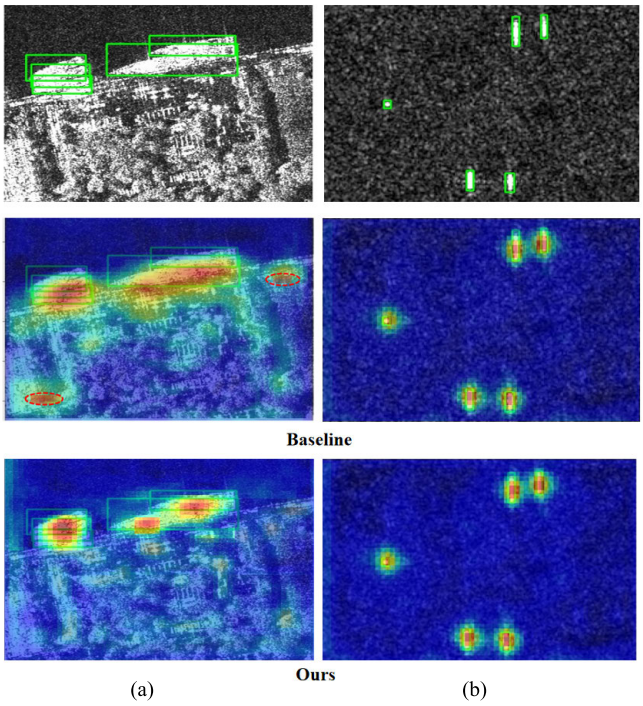


Fig. 1. First row represents the real SAR image we used in the dataset, and the green box represents ground truth. The second and third rows are feature maps comparison between the baseline and our proposed model under different background conditions. On the feature map, red and yellow represent the high response of the ship foreground, while green and blue represent the low response. (a) Complex background. (b) Simple background.

(LAE-PAFPN), which incorporates three parallel ECA attention modules to refine ship features and suppress speckle noise interference. Wang et al. [32] utilized the variance weighted information entropy (VWIE) [33] method to measure the local dissimilarity between objects and their neighborhoods, integrating a best window selection mechanism based on multiscale local contrast in VWIE calculations. Wu et al. [34] designed an efficient multiscale constant false alarm rate detector using a generalized gamma distribution [35] clutter model to detect candidate ships at sea, along with an information aggregation module to effectively integrate different feature layers of the backbone network.

At the same time, during feature extraction for dense-weak ships in complex SAR image backgrounds, convolutional operations often result in overlapping object features. This overlap can cause the object detection network to misclassify multiple densely packed ships as a single ship or even fewer ships. To address the decrease in detection accuracy in such dense areas, Sun et al. [36] proposed an anchor-free method for ship detection in SAR images, which achieves better performance by making pixel-by-pixel predictions. They applied FCOS as the base network and introduced a category-position (CP) module to optimize the position regression branch, improving target positioning in complex scenes. In addition, VfNet [37] introduced a novel loss function called varifocal loss, which emphasizes the learning of the intersection over union (IoU)-aware classification score (IACS). The network also proposed a star-shaped bounding box representation to enhance IACS prediction and refine the bounding boxes, further improving detection accuracy in dense areas.

Moreover, BiFA-YOLO [38], a YOLO-based method designed for arbitrary-oriented ship detection, addresses the challenges of detecting multiscale, arbitrarily oriented, and densely arranged ships. The method includes a bi-directional feature fusion module (Bi-DFFM) and an angular classification structure to efficiently detect ships at various orientations. Similarly, FreeAnchor, developed by Zhang et al. [39], modifies the traditional anchor allocation mechanism of anchor-based networks. Compared to RetinaNet [40], FreeAnchor demonstrates superior performance in detecting densely packed objects, reducing the effect of anchor misalignment. Furthermore, DETR with improved denoising anchorboxes (DINO), built upon the DETR [25] algorithm, introduces contrastive denoising training (CDN) and mixed query selection techniques, which are particularly effective in handling dense scenes with complex backgrounds in SAR images. These innovations enable DINO [41] to excel in detecting small objects and densely arranged targets, making it particularly suited for SAR-based ship detection tasks.

Based on the above analysis and research, it is not difficult to find that ship detection in SAR images still faces two difficulties: on the one hand, the ship detection network is susceptible to interference from complex backgrounds, and the features of the background and dense ships are often very similar, making it difficult for the network to detect ships, and the network's attention to ships is reduced, leading to a decrease in accuracy. On the other hand, during the feature extraction stage [42], the features between dense-weak ships in complex backgrounds may overlap and interfere with each other, making densely arranged ships prone to being missed or incorrectly detected by the network during the detection process.

Therefore, in response to these two problems, we use the VfNet [37] network as the baseline and design the foreground-guided background generation network (FGBG-Net) to improve the detection performance [20] of ship detection under SAR images. Here, our specific contributions are as follows.

1) We propose a Gaussian foreground localization (GFL) module that fits the localization distribution of ship foreground, identifies the approximate ship foreground regions on the feature map, provides foreground focal points for each layer of the feature map, and emphasizes features based on these positions. This module enhances the detection capability for dense-weak small ships, while alleviating the challenges posed by complex backgrounds in traditional networks.
   In addition, the GFL module utilizes foreground information to guide the removal of background features during subsequent processing stages, further reducing interference from redundant features in the model.
2) Due to the interference and feature overlap caused by complex backgrounds affecting dense and weak ships, we designed a background feature removal (BFR) module. This module uses the approximate foreground obtained from the first part to guide the network in better identifying invalid background features on the feature maps, to weaken the interference of redundant

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information on detection. Furthermore, the BFR module removes background features layer by layer, suppressing feature overlap caused by dense, weak features, thus improving the detection performance of dense and weak ships in complex backgrounds.

3) We design an FGBG-Net based on the above two modules, which utilizes the proposed GFL and BFR modules. This network improves detection accuracy, especially in dense ship scenarios within complex environments in SAR images.

The remaining parts of this article are as follows. First, Section II introduces the relevant work related to our method. Next, Section III describes the details of our proposed method. Section IV introduces the dataset and experimental settings used in this article and the corresponding experimental analysis. Section V is the conclusion of this article.

II. Related Work

A. Gaussian Mixture Model (GMM)

As a probabilistic model, the GMM [43] is usually used to model data composed of multiple fit Gaussian distributions. Each Gaussian distribution is called a component, and each data point is combined by these components with a certain weight. Next, we introduce the Gaussian distribution.

1) Gaussian Distribution: Gaussian distribution is a continuous probability distribution represented by mean and variance

$$\mathcal{N}\left(x \mid \mu, \sigma^2\right) = \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{1}$$

where $\mathcal{N}$ is the probability density function of the Gaussian model, x is the data point used for modeling, and σ and μ are the data variance and data mean. As an unimodal distribution, the Gaussian model reaches a probability peak at the mean. The variance affects the shape of the distribution, indicating the degree to which the data is concentrated around the mean. Specifically, the larger the variance, the more dispersed the data points are and the wider the distribution; the smaller the variance, the more concentrated the data points are and the denser the distribution, so the variance affects the shape of the distribution.

2) Gaussian Mixture Model: GMM consists of multiple Gaussian distributions mixed together to adapt to complex data distribution

$$\begin{cases} p(x) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}_k\left(x \mid \mu_k, \sigma_k^2\right) \\ \textbf{s.t} \sum_{k=1}^{K} \pi_k = 1 \end{cases} \tag{2}$$

where K represents the number of Gaussian components, and k represents the kth Gaussian distribution. π_k represents the distribution weight, $\mathcal{N}_k$ represents the kth Gaussian distribution. The GMM has excellent flexibility, on the one hand, it can adapt well to changes in data and dynamically update parameters with training. On the other hand, as a kind of soft clustering, each data point has a certain probability of belonging to each Gaussian distribution, rather than being strictly divided into a cluster, which makes the model more able to cope with the uncertainty in the data.

B. VarifocalNet

In recent years, object detection networks have emerged one after another, but the classification score of an object does not represent the quality of the prediction of its location, to address this problem of object detection, VarifocalNet [37] designed by Tian et al. not only has excellent detection accuracy, but also excellent in dense object detection. So we choose this network as our baseline. Next, we focus on the IACS-IoU-aware and varifocal loss parts of VarifocalNet (VfNet).

1) IACS-IoU-Aware: In the VfNet network, IACS is defined as a scalar element on the classification vector, the Ground Truth of the classification scalar is the IOU score between the predicted bounding box and the ground truth of the bounding box. This method uses appropriate scores to correctly rank a large number of candidate boxes, improving the performance of dense object detectors.

2) Varifocal Loss: Varifocal loss is designed to train dense object detectors to regress IACS

$$\mathrm{VFL}(p, q) = \begin{cases} -q(q\log(p) + (1-q)\log(1-p)), & q > 0 \\ -\alpha p^{\gamma}\log(1-p), & q = 0 \end{cases} \tag{3}$$

where p is the predicted IACS, q is the target IoU score, α and γ are the balance coefficient, for positive samples, q is the IoU between the predicted bounding box and the ground truth box, and for negative samples, q is 0. This loss function is inspired by the focal loss function, which solves the problem of imbalance between positive and negative samples while regressing the IACS score. This loss function causes the network to pay more attention to those high-quality positive samples, thereby obtaining higher AP in detection.

III. Method

Our method is mainly divided into two parts. In the first part, we design a GFL module using the GMM [43] to estimate high probability locations on the feature map that may be ship foreground, enabling us to better emphasize ship foreground features and guide remove background. In the second part, we use the Gaussian probability distribution from the first part to design a BFR module, which layers and removes background features from the classification feature map to obtain more accurate dense-weak ship detection in SAR images.

A. GFL Module

In this section, we input the image into the network and obtain the approximate foreground location of the ships through the GFL module. This module can find the rough positions of the ships' foregrounds, which may include some contextual information and background but can still preliminarily locate the foreground positions of the ships and provide enhanced probability information. This not only allows us to more accurately locate the ships in the second part but also improves the overall detection performance.

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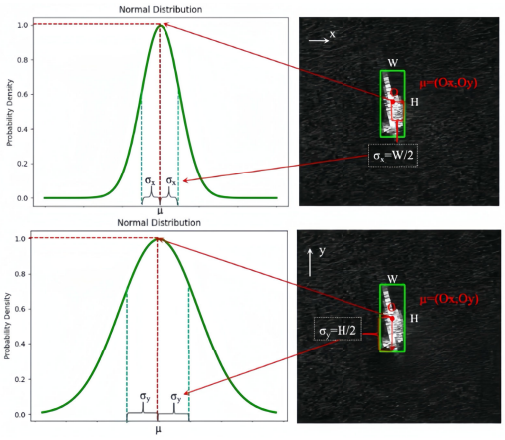


Fig. 2. Ship normal distribution in x- and y-direction in GMM. Where O represents the center of the ship bounding box, and W and H represent the width and height of the bounding box.

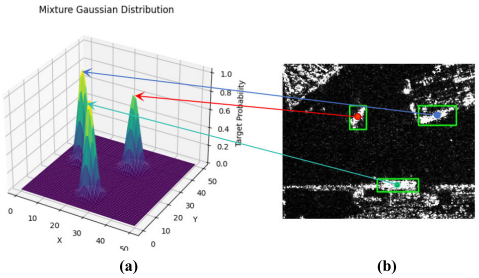


Fig. 3. Visualization results of 2-D GMM on ship detection. (a) Three-dimensional visualization result of the 2-D Gaussian model of (b), the red dots and green boxes on (b) represent the center point and bounding box of the ship, respectively.

1) Fitting of 2-D GMM to Ship Foreground: Through observation, we found that the spatial distribution of ships in SAR images aligns well with the probabilistic structure of the GMM model. Features closer to the ship's center have a higher probability of being foregrounded, while those farther away have a lower probability. This makes the GMM probability distribution an effective fit for ship localization. As shown in Figs. 2 and 3, we design the GFL module to adapt to ship foreground features by employing a 2-D GMM model, which estimates the foreground probability in both the x- and y-direction. Considering that the directional information of ships in SAR images is often ambiguous due to complex backgrounds, we simplify the modeling by adopting a vertical Gaussian distribution. This ensures robust localization while reducing the reliance on explicit orientation cues, making the model more adaptable across different SAR datasets. In addition, this symmetric distribution enhances noise resistance and prevents potential distortions caused by uncertain ship orientations.

Specifically, we need to design a GMM model that fits ship features. The following is the probability density function of the GMM model we are using:

$$\begin{aligned}\mathcal{N}\left(\mathbf{p}_{(x,y)} \mid \boldsymbol{\mu}_k, \boldsymbol{\xi}_k\right) &= \sum_{j=1}^{K} \pi_j \frac{1}{\sqrt{(2\pi)^2 \left|\xi_j\right|}} \\ &\quad \cdot \exp\left(-\frac{1}{2}\left(p_{(x,y)}-\mu_j\right)^T \xi_j^{-1}\left(p_{(x,y)}-\mu_j\right)\right) \quad (4)\end{aligned}$$

$$\boldsymbol{\xi} = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} \quad (5)$$

where π_j represents the weight of the jth GMM model as the mixing coefficient, ξ represents the covariance, μ represents the data mean, and $p_{(x,y)}$ represents any 2-D coordinates (x, y) on the feature map. K represents the number of GMM models.

In our GMM model, we utilize the Gaussian distribution to represent the probability of ship foreground. As shown in Fig. 2, we use the center O of the ship as the mean μ to represent the highest probability of the foreground and utilize the size of the ship as the covariance ξ to provide a high probability within a reasonable range. Therefore, we define the ideal μ and ξ as the center of the ship and the $H/2$ and $W/2$ of the ship's bounding box, respectively. In this case, as the distance between the feature point p_i and the center O increases, the probability of the foreground should become smaller accordingly. At the same time, we treat each ship appearing in the image as a separate Gaussian model and set the number of GMM models K to the number of ships in the image, and the mixing coefficient π_j to the ratio of the ship size. As shown in Fig. 3, we show the mixed Gaussian model fitting in the case of multiple ships.

2) Fitting of Ship-ExpectMax Algorithm: As an iterative algorithm, the expectation-maximization (EM) algorithm [43] is used to train and update the parameters of the GMM module, here we use it to update our ship foreground parameters. As shown in Algorithm 1 (SHIP-ExpectMax).

In step E, we first randomly initialize the ship parameters of the model, then calculate the feature probabilities belonging to the kth ship foreground, and then classify the feature points into different ship distributions.

In step M, after calculating the ship foreground probability of each feature point, we need to use these probabilities to update the GMM model parameters to obtain a foreground distribution closer to the ship ground truth. Therefore, when the parameter changes caused by algorithm iteration are very small, we stop iteration and finally use the loss function we designed for the GMM model to optimize ship foreground prediction.

3) Loss Function of GFL Model: After completing the update of Algorithm 1, we obtained the complete GMM model parameter, but the model at this time may not have the most ideal foreground parameters, so we need to use a loss function to optimize it

$$\begin{aligned}L_{\mathrm{GFL}} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\mathcal{N}^{\mathrm{gt}} \cdot \sum_{j=1}^{K} w_j \mathcal{N}\left(\mathbf{p}_i \mid \mu_j\right)\right) \\ + \sum_{j=1}^{K}\left\|\mu_j - \mu_j^{\mathrm{gt}}\right\|^2 + \sum_{j=1}^{K}\left\|\xi_j - \xi_j^{\mathrm{gt}}\right\|_F^2\end{aligned}$$

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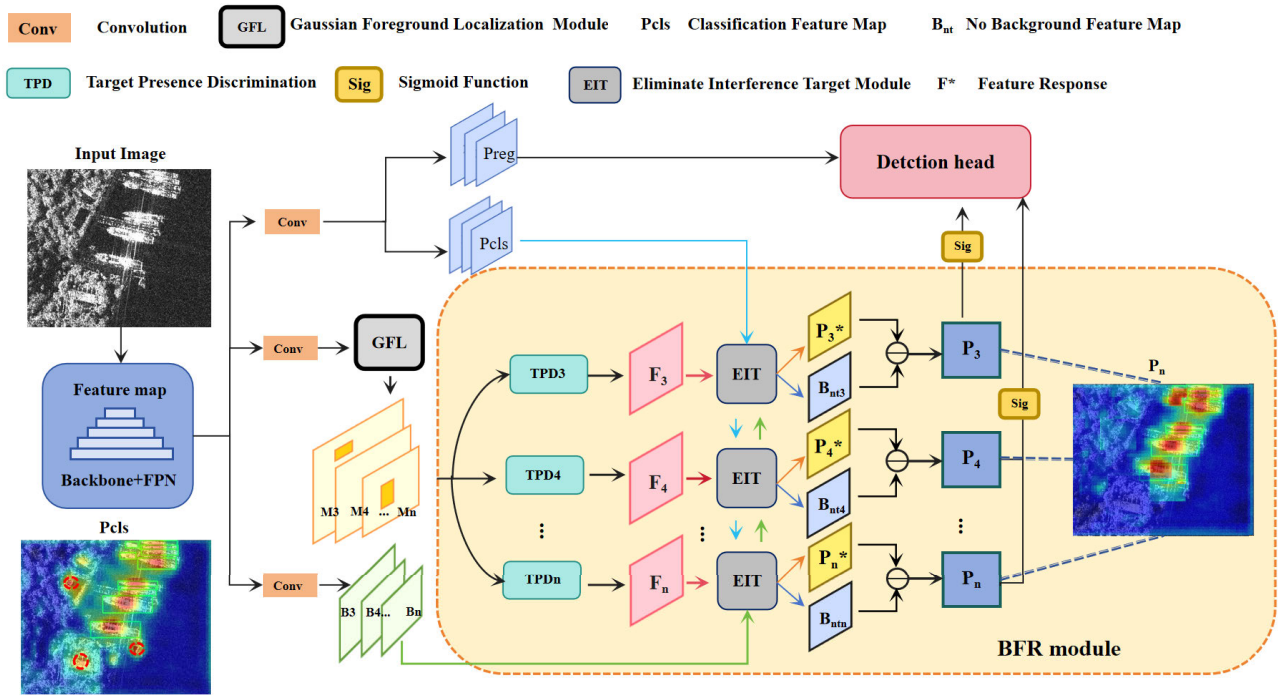


Fig. 4. Overall network structure. We input the image into backbone and FPN to obtain a series of feature maps. Through the GFL module, we obtain the foreground probability map M_n. At the same time, through convolution operation, we obtain the classification feature P_{cls} (the dashed red circle on P_{cls} represents the ship misdetected by the network) and the background feature map B_n, respectively. We send M_n, P_{cls}, and B_n to the BFR module. In BFR module, we use the target presence discrimination (TPD) block to identify the ship presence of M_n, then, in the eliminate interference targets (EITs) block, we through the block to obtain the correctly layered "pure" feature map P. Finally, the network uses Sigmoid to normalize the confidence score and sends it to the final detection. On the feature map, red and yellow represent high response of ship foreground, while green and blue represent low response, and the red dashed circle represents the ship foreground of network misjudgment.

$$\text{s.t.} \begin{cases} \mu_j^{\text{gt}} = \left\lfloor \left(\dfrac{O_{xj}}{s}, \dfrac{O_{yj}}{s} \right) \right\rfloor \\ \xi_j^{\text{gt}} = \begin{bmatrix} \left\lfloor \left(\dfrac{W_j}{2s} \right)^2 \right\rfloor & 0 \\ 0 & \left\lfloor \left(\dfrac{H_j}{2s} \right)^2 \right\rfloor \end{bmatrix} \end{cases} \tag{6}$$

where N represents the number of feature points, K represents the number of Gaussian models obtained after Algorithm 1, and μ and ξ represent the mean and variance after iteration. For the ith sample, its corresponding point is p_i, $\mathcal{N}$ is the probability density in (4), μ_j^{gt} and ξ_j^{gt} are the ground truth of μ and ξ of ship, (O_{xj}, O_{yj}) represents the center point of the jth ship, while W_j and H_j represent the width and length of the jth ship, s represents the corresponding convolution strides of the feature layer, and $\mathcal{N}^{\text{gt}}$ is the Gaussian probability calculated based on ground truth.

In (6), we use the logarithmic function and difference function to constrain our GFL module. When μ and ξ deviate from our ground truth, the probability density function multiplied by $\mathcal{N}$ decreases, causing the logarithmic function to give L_{GFL} a significant loss. At the same time, the difference function of the last two terms in L_{GFL} also guides μ and ξ in the correct direction. By designing L_{GFL}, we aim to guide the model in accurately obtaining the approximate foreground localization distribution of ships. Providing the network with locations that emphasize foreground features and suppress background features, also ensures more precise in the BFR module. Ultimately, we use L_{GFL} to optimize the GFL model, achieving ideal ship foreground localization.

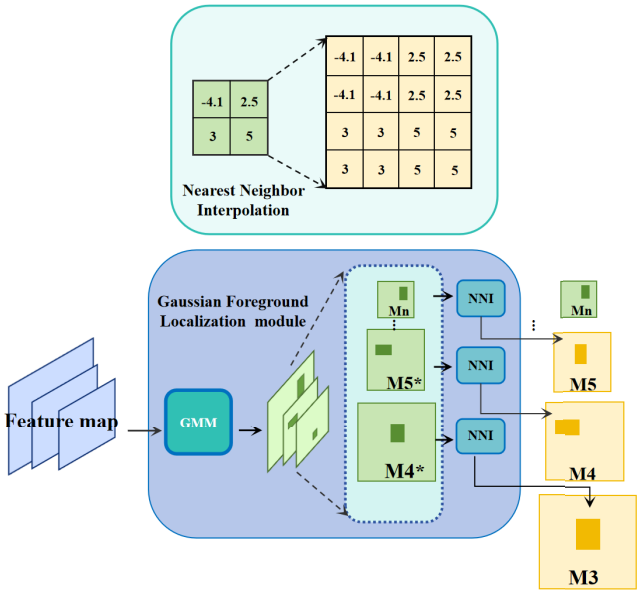


Fig. 5. Overall structure of the GFL module. GMM represents the Gaussian mixture probability model on the detection feature map. In foreground probability maps M_n, darker colors represent high probability positions of the feature map.

4) Overall Network Structure of GFL: Next, we introduce the composition of the GFL module. Fig. 4 shows the overall

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Algorithm 1 SHIP-ExpectMax

Require: 1. Initialization for each ship, a feature point is selected as the initial center. Then, initialize the μ and ξ matrix and mixing coefficients π of each Gaussian distribution.

Ensure: Completed iterative GMM model

for each iteration **do**

Expectation: E step

For each feature point, calculate the posterior probability that it belongs to the Gaussian distribution, that is, calculate the weight of each Gaussian distribution at the feature point.

$$w_{ij} = \frac{\pi_j}{\sqrt{(2\pi)^d \left|\xi_j\right|}} \exp\left(-\frac{1}{2}\left(p_i - \mu_j\right)^T \xi_j^{-1}\left(p_i - \mu_j\right)\right)$$

where p_i represents the i_{th} feature point on the feature map, μ_j represents the center coordinate of the j_{th} target. According to the posterior probability w_{ij}, each feature point i is divided into j_{th} possible Gaussian distribution.

Maximization: M step

For each Gaussian distribution, update the μ and ξ and mixing coefficient according to the feature points it is divided into.

$$\pi_j^{(t+1)} = \frac{1}{N}\sum_{i=1}^{N} w_{ij}$$

$$\mu_j^{(t+1)} = \frac{\sum_{i=1}^{N} w_{ij} p_i}{\sum_{i=1}^{N} w_{ij}}$$

$$\xi_j^{(t+1)} = \frac{\sum_{i=1}^{N} w_{ij}\left(p_i - \mu_j^{(t+1)}\right)\left(p_i - \mu_j^{(t+1)}\right)^T}{\sum_{i=1}^{N} w_{ij}}$$

Repeat steps **E** and **M** until the parameters are no longer updated and converge.

end for

structure of our network, and Fig. 5 shows the specific details of the GFL module. First, we process the feature maps using the GMM model to obtain the approximate foreground of ships across all feature map layers (as illustrated by the green feature map). Next, we utilize convolutional strides in the FPN and extract higher resolution feature (as shown by the yellow feature map) results from the previous layer's feature maps using nearest neighbor interpolation (NNI). Unlike other interpolation methods, NNI maintains the original feature scale, ensuring consistency across different feature map layers. This helps preserve ship foreground features, leading to more accurate localization.

We apply the approximate foreground probability location information obtained from the GFL module to classification feature maps for feature emphasis, assigning higher probability "rewards" to detected ship foreground positions, while providing lower probability "punishments" for interference features unrelated to ship foreground. In the subsequent BFR module, we eliminate invalid features layer by layer based on this probabilistic location information. In this way, the GFL module can effectively reduce the interference of complex backgrounds, avoid invalid features being misidentified as ship targets, and significantly improve the efficiency of ship detection.

B. BFR Module

Through the GFL module, we initially obtain a rough foreground of the ship on the feature map. However, for dense-weak ships in SAR images, the complex background not only interferes with ship detection but also shifts the network's focus toward background features, reducing attention to the ships themselves and causing feature overlap among dense ships. To address this, we design a BFR module, as shown in the overall network in Fig. 4. This module utilizes the rough foreground map M_n obtained in the first stage to guide background separation across different ships. It then removes the layered background features map B_{nt} from the classification feature map P_{cls}, effectively reducing false and missed detections in dense-weak ship scenarios.

1) TPD Block: In the BFR module in Fig. 4, we input M_n into the TPD block to detect ship presence discrimination. In the TPD block, we want to find those feature points that are within the ship's variance ξ and give them a greater response

$$\text{thre} = \frac{1}{V}\sum_{i=0}^{V}\sum_{j=1}^{K}\mathcal{N}^j(\mathbf{p}_i), \quad \text{if } D_{\text{Eu}}(\mu_j, p_i) \le \xi_j \tag{7}$$

$$F_n = \begin{cases} \dfrac{c}{D_{\text{Eu}}(\mu_j, p_i)}, & \mathcal{N}(\mathbf{p}_i) > \text{thre and } D_{\text{Eu}}(\mu_j, p_i) \le \xi_j \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where thre is the threshold we designed, V is the number of points that satisfy the if condition, K is the number of Gaussian models, p_i is the ith point on the M_n, μ_j and ξ_j are the jth mean and variance of ships on the M_n, D_{Eu} is the high response range of the jth ship, we use the Euclidean distance [44] to calculate, and c is the adjustment parameter, we obtained through experiments best results. We consider the probability mean in the range greater than thre as the ship's high response range.

In this step, we consider features greater than thre and within the range defined by μ as the center and ξ as the boundary to be the ship's foreground features. Since features closer to μ have a higher probability of belonging to the foreground, this selection helps refine the ship's foreground representation. After obtaining a blurred foreground range through the GFL module, we aim to extract a more accurate foreground using the TPD block, which further separates valid ship features from background noise in M_n. By setting a threshold thre, we classify interfering ships that do not belong to the current feature map layer as background. In addition, to reduce feature overlap and mutual interference in dense-weak ship scenarios, we treat ship features within the ξ range but below thre as background, reducing confusion between closely spaced ships. As a result, we assign a value of 0 to the background area to indicate a low feature response. This step also mitigates the

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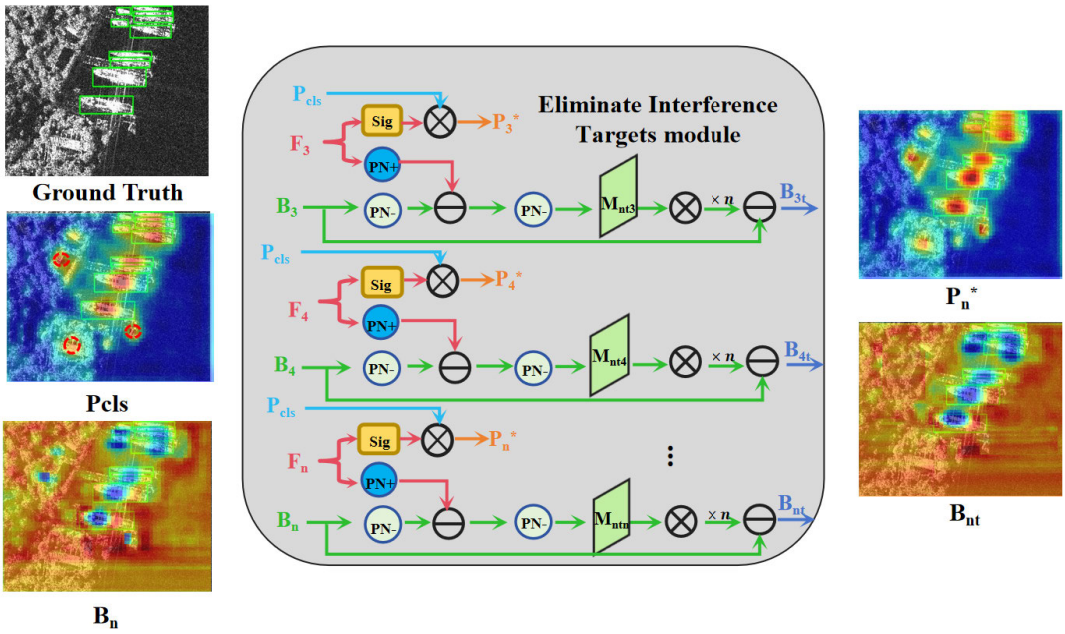


Fig. 6. EIT module. In this module, we use F_n to guide B_n to perform the correct layering. We present the feature maps generated by the network during this process, where P_{cls} and P_n^* are the target feature maps for ship, Bn and Bnt represent the background feature maps, and red represents high response, while green and blue represent low response.

blocky effect introduced by the NNI used in the GFL module, ensuring smoother feature transitions. Finally, we obtain the feature response Fn through the TPD block, which serves as one of the inputs to the next stage.

2) EITs Block: In this step, we input the classification feature Map P_{cls} and background feature map B_n, as well as the F_n obtained from the TPD block, into the EIT block as shown in Fig. 6, hoping to obtain P_n^* with feature emphasis and correctly layered background features B_{nt}

$$\begin{cases} P_{\text{cls}_{1,2,\ldots,n}} = \text{Conv}_{3\times3,1}[F_{m1}, F_{m2}, \ldots, F_{mn}] \\ B_{1,2,\ldots,n} = \text{Conv}_{3\times3,1}[F_{m1}, F_{m2}, \ldots, F_{mn}] \end{cases} \tag{9}$$

$$P_n^* = P_{\text{cls}_n} \otimes \text{Sig}(F_n) \tag{10}$$

where F_m represents the feature map shown in Fig. 4, obtained by convolution (Conv) of backbone and FPN network, and Sig$(\cdot)$ is the Sigmoid function. We obtained our P_{cls} and B_n through Conv operation. As shown in B_n in Fig. 7, we hope that the ideal B_n is opposite to P_{cls}, where the background and foreground are positive and negative numbers with high absolute values, respectively, which facilitates us to increase the difference between the foreground and background features.

First, in order to better highlight the foreground of the ship, we need to emphasize the features of P_{cls}. We use the F_n obtained in the TPD block and convert its probability to $[0.5, 1]$ using Sig$(\cdot)$ to emphasize P_{cls}, ensuring that ship features receive stronger attention while suppressing less relevant information. Then, we obtain the P_n^* emphasized by foreground features. It is evident that the more prominent the ship features are, the higher their assigned probability, which helps distinguish ships from the background. However, emphasizing only the foreground cannot effectively suppress the detection interference caused by complex backgrounds, therefore, in the next step, we further refine P_n^* by removing residual background interference layer by layer.

Second, due to the dense distribution of ships, object detection networks often struggle with small ships, as they tend to focus more on large ships. In addition, certain background regions share similarities with ship features, such as when a large ship is positioned close to a small ship, making it difficult to separate them. Therefore, in order to eliminate such misleading background features on P_n^*, we introduce a structured layering of the background feature map B_n, ensuring a more precise separation between foreground and background

$$M_{\text{nt}_n} = \left[pn^-\right]\left(B_n\left[pn^-\right] \ominus F_n\left[pn^+\right]\right) \tag{11}$$

$$\left[pn^+\right] = \begin{cases} 1, & \text{if } p^i \geq 0 \\ 0, & \text{if } p^i < 0 \end{cases} \tag{12}$$

$$B_{\text{nt}} = B_n \ominus n \times \left(B_n \otimes M_{\text{nt}_n}\right) \tag{13}$$

where $[pn^+]$ is our 0,1 discrimination on the feature map, and $[pn^-]$ is exactly opposite to $[pn^+]$, B_n is the background feature map we obtained in (9), F_n is foreground discrimination map obtained in (8), and n represents the number of feature layers.

Finally, we hope to eliminate ship features on B_n that do not belong to the nth layer feature map, as these ship features can interfere with accurate detection in dense ship under complex backgrounds, therefore, we consider them as part of the background features. To achieve this, we need first identify the ships that are causing interference. In (11), we perform a binary discrimination (0,1) on B_n and F_n, generating a mask that contains all ship foregrounds and another mask that correctly layered ship foregrounds, respectively. Then, we obtain all ship locations M_{nt} that interfere with the feature layer. As a mask, M_{nt} can help us find all ship locations that

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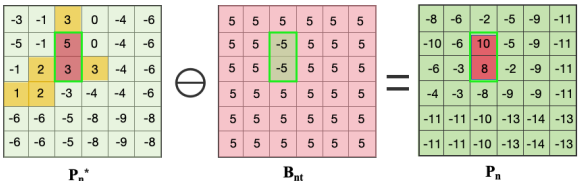


Fig. 7. Formation process of P_n. The red boxes inside the green boxes represent our ship's position, while the surrounding yellow boxes represent interference features formed by dense ships or complex backgrounds.

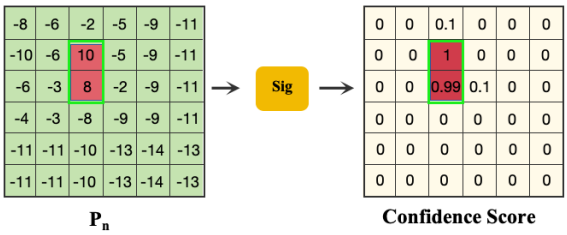


Fig. 8. Process of forming classification score from P_n.

interfere with detection on B_n. We subtract the features at these locations from B_n to ensure that they successfully change from the erroneous ship foreground to part of the background. Ultimately, this process yields background features B_{nt} that exclusively contain correctly detected ships.

3) Classification Background Layer Removal: To further enhance ship features and suppress interference, we optimize the classification feature map by eliminating residual background redundancy. By removing incorrect ship features that do not belong to the current feature layer, we aim to enable the network to focus more precisely on the ship foreground, thereby improving detection accuracy. The process is formulated as follows:

$$P_n = P_n^* \ominus B_{\text{nt}} \tag{14}$$

where n represents the nth feature layer. As shown in Fig. 7, this operation removes background interference from P_n^*, resulting in a cleaner feature representation with stronger foreground emphasis. This refinement prevents the network from being distracted by irrelevant features, allowing it to focus more effectively on ships.

In addition, by leveraging our layered ship feature extraction strategy, we mitigate feature overlap and mutual interference in densely packed ship scenarios. This enhances the network's ability to differentiate ships in complex maritime environments, ultimately leading to more accurate detection.

Finally, as shown in Fig. 8, the refined feature map P_n is passed through the Sig(·) function to obtain the final classification score, completing the detection process with improved precision.

C. Loss Function

To optimize the background map and enhance feature separation, we introduce a loss function that guides the network to differentiate ship and nonship regions more effectively. By aligning the predicted background features with the ground truth, this loss function encourages clearer foreground–background distinction, reducing false detections. The loss function is defined as follows:

$$L_B = n \sum_{t=0}^{n} \frac{1}{N} \left(1 - \frac{\text{Sig}\left(B_t\left(\sum_{k,h} p_{kh}\right) \cdot g_{kh}\right) + \epsilon}{\text{Sig}\left(B_t\left(\sum_{k,h}(|p_{kh}| + |g_{kh}|)\right)\right) + \epsilon} \right) \tag{15}$$

where n is the total number of feature layers, N represents the number of feature points, and B_t denotes the background map at the tth layer. The terms p_{kh} and g_{kh} correspond to the values at the kth row and hth column of B_t in the predicted map and the ground truth, respectively.

We define the ground truth of B_t such that feature points at ship locations have a value of $-n$, while other feature points are assigned a value of n. The background map is then passed through Sig(·), after which the loss is computed. L_B ensures that p_{kh} is trained to approach g_{kh}, enhance the responsiveness of the ship foreground and suppress irrelevant background features.

When there is a significant discrepancy between p_{kh} and g_{kh}, particularly in cases where one is positive and the other is negative, L_B produces a higher loss to penalize the error. Conversely, when the values have the same sign and their difference is small, the loss is reduced. To prevent division by zero, a small constant ϵ is added to both the numerator and denominator, ensuring numerical stability during training.

The overall loss function of the whole method is as follows:

$$\text{Loss} = \text{VFL}(p_c, q_c) + \frac{\alpha}{\text{N}_{\text{pos}}} \sum^{N} L_{\text{GFL}} + \frac{\beta}{\text{N}_{\text{pos}}} \sum^{N} L_B \tag{16}$$

where VFL(·, ·) is the original loss function in the VfNet [37], we introduced it in (3), p_c and q_c are the predicted value and Ground Truth of category c, respectively, N_{pos} is the number of positive samples detected, α and β is the loss balance coefficient. In this article, we follow the parameters of the original paper for all parameters in VfNet.

IV. Experiments

A. Dataset Description

1) SSDD: This dataset [6] is a ship detection dataset for SAR, with an average image size of 500 × 500. It contains multiple imaging results [45], including RadarSat-2, TerraSAR-X, and Sentinel-1 sensors, as well as four polarization methods, HH, HV, VV, and VH. This dataset contains 1160 images, with a total of 2456 ships, with an average of 2.12 ships per image. In addition, ships in this dataset have different resolutions of 1–15 m, ranging from inshore to offshore. The range of ship sizes from small to large is 7 × 7 to 211 × 298. In this dataset, most of the ships are small objects. In this article, we divide the SSDD dataset into training, validation, and test sets in a 7:2:1 ratio.

2) HRSID: This dataset [46] is composed of high-resolution SAR images for ship detection, semantic segmentation, and instance segmentation tasks. The image size is 800 × 800. This dataset includes 5604 high-resolution SAR images with 16951 ship instances. The original images are 99 Sentinel-1B

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images, 36-TerraSAR-X, and 1-TanDEM-X images. HRSID dataset with resolutions of 0.5, 1, and 3 m also draws on the construction process of the COCO dataset. The dataset is divided into a training set and a test set in a ratio of 13:7.

B. Evaluation Indicators

1) Average Precision: In this article, we use AP as our evaluation metric, so we first need to introduce two important performance metrics for object detection models: Precision in (17) and recall in (18)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

where true positive (TP) is the number of samples correctly predicted as positive by the model. False positive (FP) is the number of samples that the model incorrectly predicts as positive. True negative (TN) is the number of samples correctly predicted as negative by the model. False negative (FN) is the number of samples that the model incorrectly predicts as negative.

Therefore, from the above concepts, it is not difficult to see that accuracy represents the proportion of detected positive targets that are actually positive targets. The recall rate represents the proportion of all actual positive targets that have been successfully detected as positive targets.

AP is a comprehensive evaluation index for the precision and recall of different classes, representing the average accuracy of a specific class. For each class, AP uses different thresholds to calculate different precision recall points, interpolates these points, and calculates the area under the curve to obtain the AP value.

AP can also be divided into APs, APm, and APl, where s represents small targets and m and l represent medium and large targets. These indicators help researchers understand the performance of the model when dealing with targets of different scales. The size division is determined based on the objectives of the COCO dataset.

2) Mean AP (mAP): mAP is a commonly used metric to evaluate the overall performance of an object detector across multiple categories, typically used to measure the overall performance of the entire model. In this article, for the single class object detection task of ship, we use mAP range in [0.5,0.95], which represents the average AP value of detecting ship within the IoU threshold range, ranging from 0.5 to 0.95 with intervals of 0.05.

C. Experimental Setup

The specific experimental parameters are set as follows.

All the experimental environments are implemented in GeForce GTX1080 11-GB GPU and 64-GB RAM under 18.04.1-Ubuntu. The computer and deep-learning training parameters for our experiments are presented in Table I. For two datasets, our models train a total of 60 epochs, which are then divided by a factor of 10 at the 40th. In order to save costs, we resize the HRSID dataset input images to 400×400 pixels. ϵ in (15), in this article, we set it to 0.0001. For the definition of target size, we follow COCO [20] absolute size definition method.

TABLE I
TRAINING PARAMETERS

Parameter	Value
Optimizer Function	SGD
Learning Rate	0.001
Momentum	0.9
Batch Sizes	8
Weight Decay	0.0001
Totally Epochs	60
Warmup Iters	500

TABLE II
MAP RESULTS OF DIFFERENT HYPERPARAMETER c ON TWO DATASET

c	SSDD	HRSID
2	0.659	0.515
4	**0.669**	0.518
6	0.664	**0.524**
8	0.662	0.520
10	0.661	0.516

D. Analysis and Comparison of Hyperparameters

In this section, we conduct a detailed analysis and comparison of the hyperparameters in this article: adjustment parameter c in (8), and loss balance cofficient α and β in (16). We adopt the AP [20] to evaluate the ship detection performance.

1) Hyperparameter Analysis for Adjustment Parameter c: It is not difficult to see from (8), we need to perform ship existence discrimination from the foreground probability map M_n to form the discrimination map F_n, without using c, the range of F_n belongs to [0, 1], while the range of the Sigmoid function in [0, 1] is only [0.5, 0.73]. This range cannot distinguish the importing feature and background reasonably, so we use the adjustment parameter c to adjust it. Table II shows our c experimental results on the SSDD and HRSID datasets, respectively.

From the experimental results, we can easily see that on the SSDD dataset, when c is 4, the best mAP is 66.9%, and on the HRSID dataset, when c is 6, the best mAP is 52.4%. This is because when c is at its optimal value, it not only emphasizes the ship features, but also suppresses the background that is not conducive to detection. At the same time, c can distinguish the features of the background and ship very well, which provides us with assistance in removing background features in the future.

2) Loss Balance Cofficient α and β: α and β are the balance coefficients in (16). We conducted experiments on two datasets. The experimental results are shown in Tables III and IV. It is not difficult to see from Tables III and IV that on the SSDD dataset, we achieved the best mAP when α and β were 1 and 1, and on the HRSID dataset, we achieved the best mAP when α and β were 1.5 and 1.

We can easily observe that due to the different sample sizes contained in the SSDD dataset and the HRSID dataset, there

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TABLE III
MAP RESULTS OF DIFFERENT HYPERPARAMETER α AND β ON SSDD

α	β	**mAP**
α = 0.5	β=1	0.662
α = 1	β=1	**0.669**
α = 1.5	β=1	0.665
α = 1	β=0.5	0.664
α = 1	β=1	0.669
α = 1	β=1.5	0.660

TABLE IV
MAP RESULTS OF DIFFERENT HYPERPARAMETER α AND β ON HRSID

α	β	**mAP**
α = 0.5	β=1	0.520
α = 1	β=1	0.521
α = 1.5	β=1	**0.524**
α = 1.5	β=0.5	0.518
α = 1.5	β=1	0.524
α = 1.5	β=1.5	0.522

are more samples that need to be trained and detected in the HRSID dataset than in SSDD. Therefore, the α value in the HRSID dataset should be slightly greater than the β value to achieve accurate positioning balance and better remove background features in the BFR stage.

E. Module Comparison Experiments

In this section, we use both two datasets to verify the effectiveness of our network, which includes GFL and BFR. We add each module to the baseline VarifocalNet (VfNet) [37] network we use. Our AP of the strategy and VfNet network applied to the corresponding dataset are recorded in Tables V and VI, and Fig. 9 shows the detection results of our network and baseline with Ground Truth. Fig. 10 shows the visualization comparison of feature maps between our network and baseline. Figs. 11 and 12 show the feature visualization results of the GFL module and BFR module, respectively. In addition, in both datasets, the improvement is relatively significant due to the smaller number of dense large ships compared to dense small and medium-sized ships.

1) Validation of the GFL: In this section, we verify the effectiveness of the proposed GFL module, as shown in Tables V and VI, VfNet∪GFL exhibits a 2.1% improvement in mAP on the SSDD dataset compared to baseline, and a 2.2% increase on the HRSID dataset.

It is not difficult to see from the results of the two datasets that, with the help of the GFL module, we have found the approximate high-probability positions of ship features on the feature map and used these high probabilities to emphasize the features of ships. Therefore, the AP of different sizes on both datasets has improved. For small ships, APs increased by 1.3% in the SSDD and HRSID datasets, on medium-sized ships by 2.8% and 3.3%, respectively, and on large ships by 2.1% and 4.8%, respectively. At the same time, as shown in Fig. 9(a), the baseline in the third and fourth row mistakenly identifies the background similar to the target as the ship, resulting in false detections. As shown in Fig. 11, with the help of the GFL module, we emphasize ship features while suppressing complex background features making ship features more prominent on the feature map, while the background is exactly the opposite, but without the GFL module.

TABLE V
EXPERIMENTAL VERIFICATION OF THE PROPOSED METHOD ON SSDD

	APs	APm	APl	mAP
VfNet	0.566	0.748	0.934	0.635
VfNet∪GFL	0.579	0.776	0.955	0.656
VfNet∪BFR	0.577	0.766	0.937	0.650
VfNet∪GFL∪BFR	**0.598**	**0.780**	**0.966**	**0.669**

TABLE VI
EXPERIMENTAL VERIFICATION OF THE PROPOSED METHOD ON HRSID

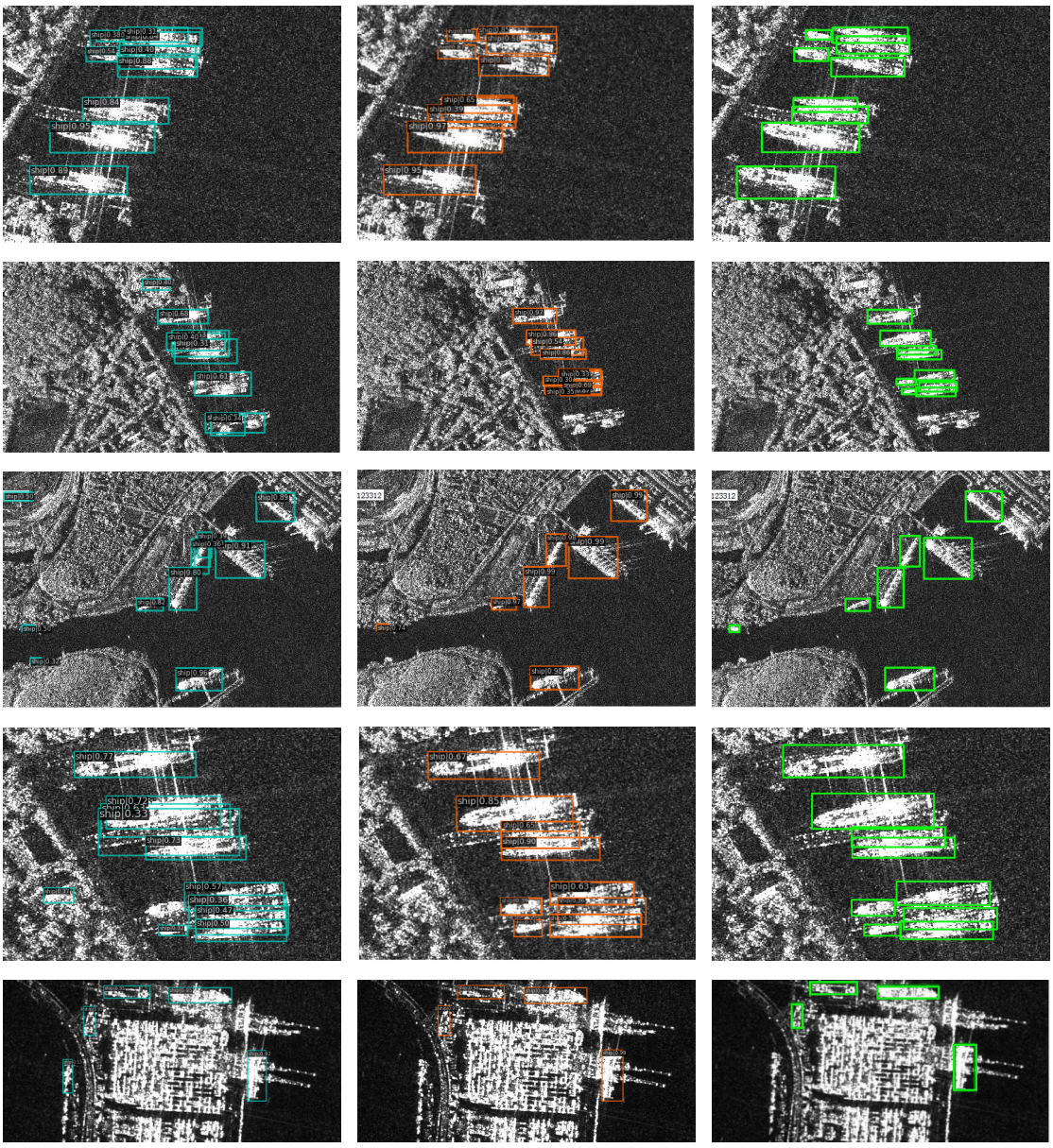
	APs	APm	APl	mAP
VfNet	0.496	0.635	0.370	0.490
VfNet∪GFL	0.509	0.668	0.418	0.512
VfNet∪BFR	0.507	0.644	0.400	0.508
VfNet∪GFL∪BFR	**0.524**	**0.670**	**0.423**	**0.524**

2) Validation of the BFR: In this section, we verify the effectiveness of the proposed BFR module, as shown in Tables V and VI, VfNet∪BFR exhibits a 1.5% improvement in mAP on the SSDD dataset compared to baseline, and a 1.8% increase on the HRSID dataset. In the BFR module, we remove all backgrounds unrelated to ships from the feature map, and use these features for detection after removing interfering backgrounds. Therefore, the mAP on both datasets has improved. For the SSDD dataset, APs increased by 1.1%, and APm increased by 1.8%. On the HRSID dataset, APs increased by 1.1%, APm increased by 0.9%, and APl increased by 3%.

The BFR module removes interference background from all ships without layering, so the improvement in AP is not as significant as VfNet∪GFL∪BFR. For the features of the ship, on the two datasets, VfNet∪GFL∪BFR increased by 3.4%, APs increased by 3.2% and 2.8%, respectively, and APm increased by 3.2% and 3.5%, respectively. APl increased by 3.2% and 5.3%, respectively. As shown in Fig. 12, we use the BFR module to eliminate all feature backgrounds unrelated to the ship, avoiding background interference in Fig. 12(a) and misjudgment caused by the complex background of the port shore in the baseline.

For VfNet∪GFL∪BFR, guided by the foreground probability in the GFL module, BFR removes all background interference, as shown in the second line of Fig. 9, (a) baseline identifies multiple dense-weak ships as one, which is caused by feature overlap, but in (b) ours, the BFR module removes the background features of dense ships layer by layer, so the network can correctly detect ships. And as shown in Fig. 10, in the first and second rows, the feature maps of the baseline model tend to misclassify complex backgrounds with features similar to ships as ships themselves, while also being more prone to missing small ships in densely arranged scenarios. In contrast, our network demonstrates a better ability to address these challenges. Furthermore, in the feature maps of the third and fourth rows, our network shows certain advantages in detecting small targets against complex backgrounds, effectively highlighting these small ships without interference from the background.

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(a) (b) (c)

Fig. 9. Detection results comparison between the baseline network and the proposed network on two datasets. The numbers on the boxes of different colors represent the confidence. (a) Baseline. (b) Ours. (c) GT.

TABLE VII

COMPARISON OF SHIP DETECTION WITH OTHER METHODS ON SSDD

	AP0.5	AP0.75	APs	APm	APl	mAP
FCOS	0.892	0.678	0.497	0.724	0.811	0.584
Deformalbe-Detr	0.952	0.749	0.561	0.728	0.900	0.623
Free-Anchor	0.940	0.761	0.576	0.737	0.855	0.628
VfNet	0.934	0.763	0.566	0.748	0.934	0.635
YOLOX-s	0.935	0.777	0.584	0.737	0.817	0.642
Cascade-RCNN	0.944	0.768	0.577	0.746	0.933	0.643
DION	0.953	0.772	0.577	0.748	0.933	0.644
DiffusionDet	**0.958**	0.774	0.579	0.718	0.900	0.649
Ours	0.955	**0.801**	**0.598**	**0.780**	**0.966**	**0.669**

F. Comparison With Other Methods

We compare it with several existing methods to quantitatively evaluate our proposed method, including FCOS [9], Deformable-DETR [24], Free-Anchor [39], VfNet [37], YOLOX-s [29], and Cascade-RCNN [15], DINO [41], DiffusionDet [47]. Tables VII and VIII, respectively, show our comparative results on two datasets, using AP and IoU with different sizes as our indicators.

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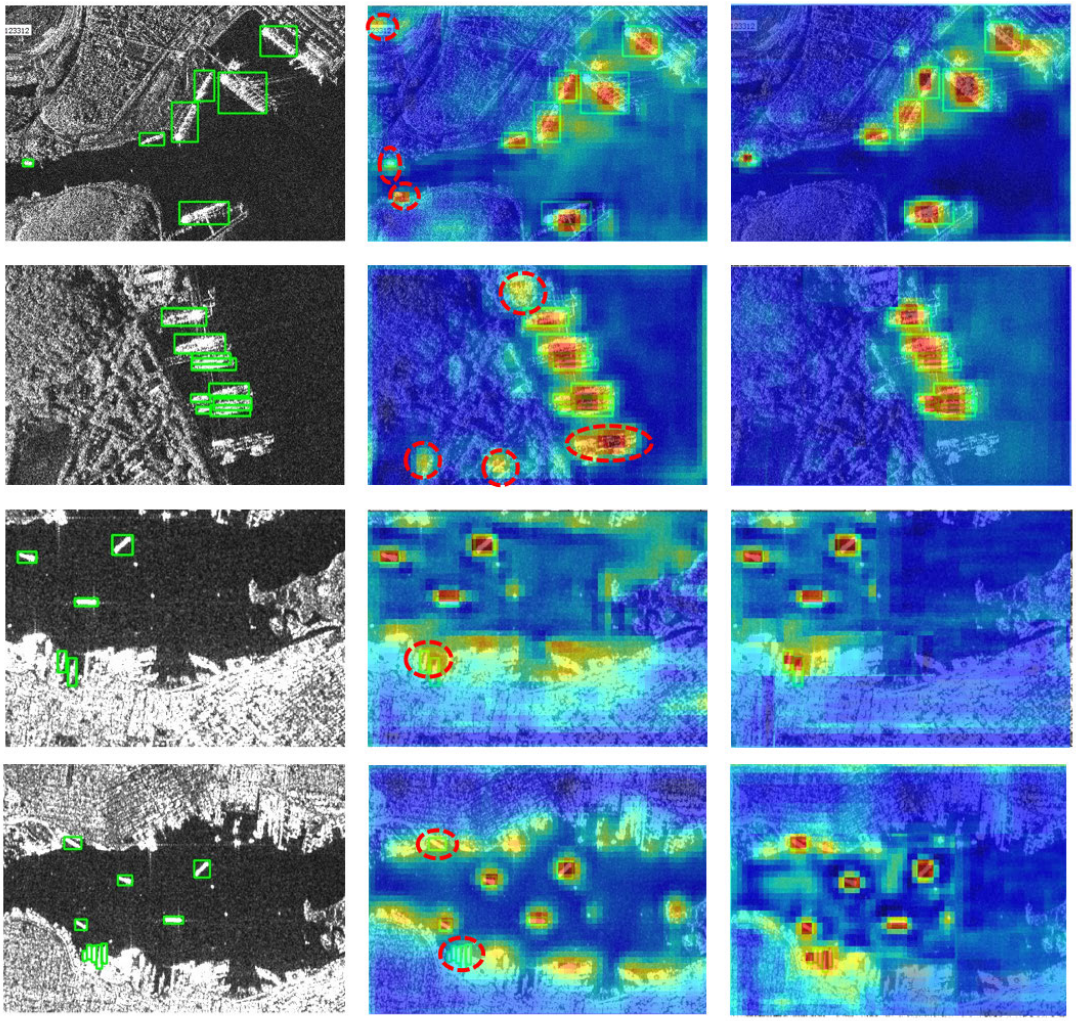


Fig. 10. Results of the visual features of dense-weak ships in different scenes, where the red part represents the high response position of the ship, the green and blue parts represent the low response positions of the ship, and the red dashed circle represents the baseline misjudgment and missing ship. (a) Ground truth. (b) Baseline. (c) Ours.

TABLE VIII

COMPARISON OF SHIP DETECTION WITH OTHER METHODS ON HRSID

	AP0.5	AP0.75	APs	APm	APl	mAP
Deformalbe-Detr	0.706	0.488	0.435	0.507	0.225	0.433
FCOS	0.720	0.522	0.476	0.653	0.298	0.466
Cascade-RCNN	0.661	0.556	0.480	0.613	0.228	0.482
VfNet	0.720	0.556	0.496	0.635	0.370	0.490
YOLOX-s	0.733	0.555	0.498	0.629	0.355	0.493
DION	0.820	0.559	0.504	0.478	0.300	0.499
Free-Anchor	0.738	0.566	0.508	0.648	0.394	0.504
DiffusionDet	0.737	0.572	0.503	0.624	0.250	0.500
Ours	**0.752**	**0.580**	**0.524**	**0.670**	**0.423**	**0.524**

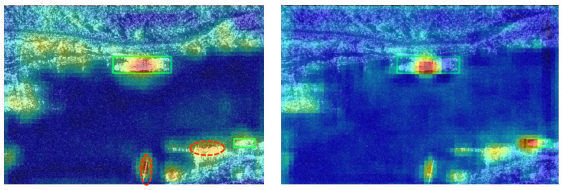
TABLE IX

COMPARISON OF INFERENCE SPEED AND COMPUTATIONAL COMPLEXITY

	Deformable-DETR	FCOS	Cascade R-CNN	VfNet	YOLOx-s	DION	Free-Anchor	DiffusionDet	Ours
FPS (img/s)	4.9	34.4	**37.5**	24.3	32.6	29.3	32.6	19.0	33.8
PIT (ms)	203.5	29.1	**26.6**	38.7	30.7	34.1	30.6	48.1	29.6
FLOPs (G)	36.558	36.816	77.301	35.398	**8.524**	54.933	37.485	141.0	41.532
Params (M)	40.119	32.295	69.395	32.892	**8.938**	47.463	36.330	35.5	38.958

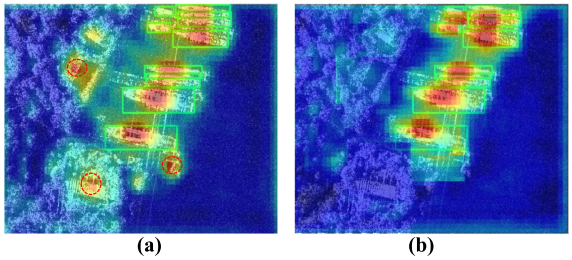
FCOS [9] network is an anchor-free lightweight network, it is excellent in terms of detection and training speed. However, at the same time, FCOS is difficult to adapt to targets with excessive scale changes. The positive and negative

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(a) (b)

Fig. 11. Validity verification of the GFL module. (a) Represents the feature map of the baseline, (b) represents our feature map, and the red dotted ellipse represents the erroneous detection ship caused by the complex background. The red on the feature map represents the high-probability location where the ship exists, the green and blue are low-probability locations, and the green boxes in the background is ground truth.



(a) (b)

Fig. 12. Validity verification of the BFR module. (a) Is the baseline feature map, (b) is our feature map, and the meaning of the colors on the map is consistent with Fig. 11.

samples of large targets and small targets are unbalanced. Especially for small targets, there are too few positive samples, which makes them easily interfered with by complex backgrounds, so the network still has room for improvement.

Deformable-DETR [24], as a transformer-based object detection network, is built on the DETR [25] network. It also uses multiscale feature detection, which can adapt well to multiscale targets in the dataset. However, we found that the bipartite graph matching used in the "postprocessing" stage of the network cannot truly remove nontarget interference background, and the detection is also affected to a certain extent by the background, so there is still room for improvement.

Free-Anchor [39] designs an optimization method from the perspective of maximum relief estimation, allowing the network to select target-matching anchors more flexibly, define detection customization likelihood, and implement an end-to-end detection and classification training mechanism to maximize the likelihood. Although it can promote the network to learn how to match the optimal anchor and ensure compatibility with the NMS algorithm [48], the matching mechanism is not flexible enough, making the FreeAnchor network inaccurate in SAR detection on dense ships.

VfNet [37] is the baseline we use. As an anchor-free network, it introduces an IOU sensing mechanism to position the target more accurately and improve the accuracy of detection. At the same time, VfNet can detect multiple targets in the image, which helps to improve the efficiency and comprehensiveness of target detection. However, VfNet's introduction of the IoU awareness mechanism and dense detection may increase computing costs and over-reliance on the IOU mechanism, therefore, there is still room for improvement in dense ship detection.

YOLOX [29] is an object detection algorithm based on the YOLO series of algorithms. Among them, YOLOX-s is a lightweight version of the YOLOX framework. It greatly reduces the calculation and complexity of the model while maintaining high detection accuracy. However, YOLOX-s's detection accuracy for small ships is not accurate enough, and there is also a lot of room for improvement in dense ship detection.

Cascade-RCNN [15], as the object detection network of the R-CNN series, effectively improves the accuracy of the detector through cascading and gradually filtering and refining. Each cascading stage can solve detection problems of different difficulty, thereby improving the overall detection effect. However, due to the cascading structure, Cascade-RCNN has a relatively slow inference speed, and it still relies on prior knowledge to set anchors, so it is still inevitably affected by background features.

DINO [41], as an object detection algorithm developed in recent years based on DETR [25], retains the simplicity of DETR. It also achieves significant improvements in both accuracy and training efficiency. However, in dense scenes, especially in maritime ship detection, complex backgrounds, and noise can affect the detection results. Although DINO introduces CDN to enhance the suppression of background interference, the model may still misdetect background as ships when dealing with highly complex backgrounds, leading to a higher false positive rate.

DiffusionDet [47] is a newly developed object detection framework that utilizes the denoising diffusion process to transform noisy bounding boxes into accurate detections. This method offers flexibility, adapting to dynamic numbers of bounding boxes, and demonstrates exceptional performance in scenes with varying object densities. When applied to weak-density ship detection in SAR images, DiffusionDet effectively optimizes noisy proposals. However, its reliance on Gaussian noise processing may lead to misclassifications in complex maritime environments, particularly due to wave interference and clutter. In addition, the iterative optimization process requires multiple iterations, increasing computational demands and potentially impacting real-time performance.

In addition, as shown in Table IX, we comprehensively evaluated the speed and computational efficiency of our algorithm using frames per second (FPS) and per image time (PIT), and compared the complexity of floating point operations (FLOPs) and Params [49] with other advanced networks.

From Table IX, our method achieves a competitive FPS of 33.8 and PIT of 29.6 ms, outperforming Deformable-DETR (4.9 FPS, 203.5 ms) and DiffusionDet (19.0 FPS, 48.1 ms) in inference speed. Compared to FCOS (34.4 FPS, 29.1 ms) and Cascade R-CNN (37.5 FPS, 26.6 ms), it maintains similar speed while achieving higher accuracy for dense and small targets. In addition, our method is more efficient, with 41.532 G FLOPs and 38.958 M parameters, compared to Cascade R-CNN (77.301 G FLOPs, 69.395 M) and DINO (54.933 G FLOPs, 47.463 M). It is not difficult to see that our algorithm can maintain superior accuracy while maintaining computational efficiency and complexity that are comparable or even superior to most algorithms, making it

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TABLE X
EXPERIMENTAL RESULTS OF EACH COMPONENT ON SSDD

Baseline	GFL	BFR	APs	APm	APl	mAP
✓			0.566	0.748	0.934	0.635
✓	✓		0.579	0.776	0.955	0.656
✓	✓	✓	**0.598**	**0.780**	**0.966**	**0.669**

TABLE XI
EXPERIMENTAL RESULTS OF EACH COMPONENT ON HRSID

Baseline	GFL	BFR	APs	APm	APl	mAP
✓			0.496	0.635	0.370	0.490
✓	✓		0.509	0.668	0.418	0.512
✓	✓	✓	**0.524**	**0.670**	**0.423**	**0.524**

a robust and practical solution for SAR-based ship detection tasks.

G. Ablation Study

In this section, we gradually add GFL and BFR to the baseline to verify the effectiveness of each part of our network. Tables X and XI show us the ablation experimental results of the two datasets, respectively.

1) GFL: In Tables X and XI, we used the GFL module to increase mAP by 2.1% in the SSDD dataset and by 2.2% in the HRSID dataset. The GFL module found the foreground of the ship for us and emphasized the features of the ship, so the mAP has improved.

2) GFL∪BFR: After we added the BFR module to the GFL module, we eliminated the interference background on the classification feature map. On the SSDD and HRSID datasets, mAP increased by 3.4%, and APs increased by 3.2% and 2.8%, respectively. APm increased by 3.2% and 3.5%, respectively, and APl increased by 3.2% and 5.3%, respectively.

Furthermore, it is worth noting that in the HRSID dataset, the AP of large ships is lower than that of small and medium-sized ships. This is mainly due to the uneven size and quantity of the dataset, with the proportion of large ships being much lower than that of small and medium-sized ships, which affects model learning.

V. CONCLUSION

In this article, our FGBG-Net uses GFL to find the approximate foreground of the ship on the feature map to find the approximate foreground of ships on the feature map for feature emphasis and background generation guidance, and in the BFR module, we remove interference background on the classification feature map, improving the detection performance of dense-weak ship in SAR images under complex background. We have confirmed the effectiveness of our network through experiments. However, there is still room for improvement and enhancement in the detection of small ships in this network, due to the size difference between large and small ships, there is still room for improvement in weight balance design in Gaussian model-based localization. In future work, we will design a ship detection network that can address the weight differences between large and small ships to improve the accuracy and generalization ability of SAR ship detection networks.

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